

Networks and the Internet

Computer Network

When two or more computers are connected to each other using a medium (e.g cables) they form a computer network

Advantages of computer network

- Can share hardware like printer, hard disks etc
- Can send E-mail
- Can share Application
- Central backup can be taken
- Data can be shared

Disadvantages of computer network

- viruses spread faster over computer network
- fear of un-authorized access(hacking)
- initial cost is high

Types of Network

Server based Network

In a server based network the whole networking/ routing is done through a host computer known as a server

Peer to peer

In this network there is no dedicated server each computer in a network acts as a server or client at the same time

Server: A computer that serves the request

Client: That does the request

Medium: Through which signal travels

WLAN	LAN	WAN
Wireless local area network	Stands for local area network	Wide area network
Can be formed in a room, building or a vicinity	It is a small network of computers that can be formed in a room, building or a vicinity	A huge network that spans from continent to continent
no wires or cables involved cheaper as compare to LAN	Medium used is cable i.e. UTP(unshielded twisted pair cable), co-axial and fiber optics	Medium used is telephone lines, satellite and fiber optics
Ethernet card with wireless capability	Device that is used to connect computers to the LAN is Ethernet card/Network interface card	Devices that connect computer to the WAN is Modem
Slower than LAN	LAN is faster than WAN	It is slower than LAN
Is less secure as compare to LAN	More secure than WAN and WLAN	Less secure as compare to LAN

Network Topology

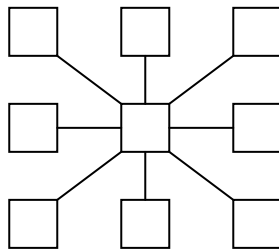
These are Methods or techniques used to form a computer network structure

Types of Topologies

- Star Topology
- Ring Topology
- Bus Topology
- Hybrid Topology

Star Topology

In a star topology computer are connected to each other through a host computer using a special device known as HUB



Advantages

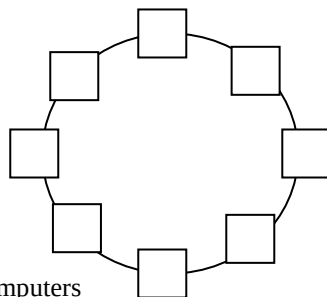
- Faster data transmission
- More secure
- If any system stops working the whole network will not be disturbed
- can add more devices easily

Disadvantages

- Expensive to install
- Dependency over host computer i.e. if host computer stops working the whole network disturbs

Ring Topology

In Ring topology each computer is connected to its two adjacent computers to form a logical ring



Advantages

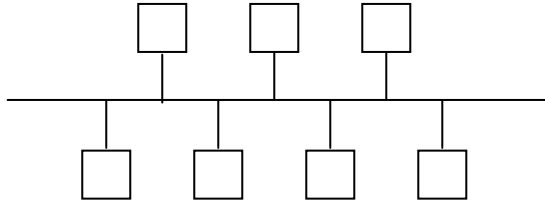
- Inexpensive to install
- Suitable to connect Less no of computers
- Faster than bus topology

Disadvantages

- If any device stops working the whole network will be disturbed
- Not reliable
- difficult to add more devices

Bus Topology

In Bus topology all computers are connected to each other through a main cable known as BUS



Advantages

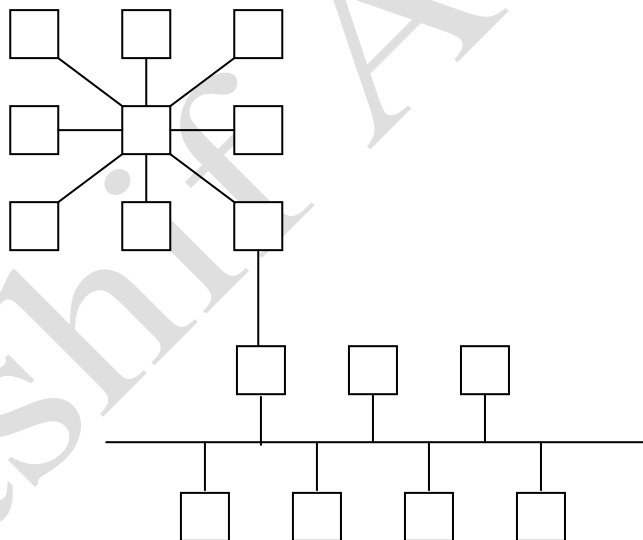
- Inexpensive to install
- Suitable to connect Less no of computers
- If any computer stops working the network is not disturbed

Disadvantages

- If BUS stops working the whole network will be disturbed
- Slower data transmission as compare to Ring network

Hybrid Topology

Combination of two different topologies



Network Devices

Following are the most commonly used network devices

- Modem
- HUB
- Switch
- Router
- Bridge
- Proxy Server
- Ethernet Card/Network Interface card

Modem:

It connects computers to the internet through telephone lines. It converts analog signals to digital (demodulation) and digital to analog signal (modulation)

Note: two devices that connect LAN to WAN are router and modem
two devices that connect LAN to internet are modem and router

Router:

It connects two dissimilar networks having same protocol. It finds and allocates the fastest route for data transmission

Note: two devices that connect LAN to WAN are router and modem
two devices that connect LAN to internet are modem and router

HUB:

It is used to connect several computers together to form a LAN. It broad casts data over network. It is not reliable as does not manage data efficiently over network

Switch:

It is also used to connect computers together to form a LAN. It sends data directly to the specified computer. Hence it is more reliable and manages data efficiently over computer network.

Proxy Server:

It is a computer system or computer application that provides users with the data or information from another network or server.

Features of proxy server

- It can act as a webserver
- It can act as a buffer (a buffer is a temporary memory used to compensate between the low speed and high speed devices)
- It can block certain websites (as fire-wall is the major component of a proxy server)
- It connects computers of network to internet
- It directs user's request to the internet

Ethernet or network interface card:

It is used to connect computer to the LAN (Local Area Network)

Bridge:

It connects two similar LANs (Local Area Network), it has two ports

Video Conference

It is a technique by which people far away from each other can have video, sound and text conversation. It is done using a broad band internet connection.

Hardware used in video conference:

- Monitor
- Webcam
- Modem
- Router
- Microphone
- Speaker
- Codec(software)

Advantages of video conference

- + No need to travel
- + Saves time to travel
- + Saves money to travel
- + No need to pay for conference room
- + Less crowded
- + Meetings can be called at short notice
- + Can edit documents electronically
- + People can work from home

Disadvantages of video conference

- + Equipment is expensive
- + Need broadband internet connection
- + Video or sound may not synchronize
- + Time zone would be a problem
- + Cannot sign agreements
- + People may feel isolated

Internet	Intranet
• It is a permanent network of networks that connects millions of computers together.	• It is a privately maintained computer network created using <u>world wide web</u> software. It is used by several organizations as a means of communication
• Less secure	• More secure
• Slower than intranet	• Faster
• Public Network	• Private network

Wiki	Blog
• It is a website that allows people to create their own pages • allow them to add and edit comments	• It's a personal diary. • That allows people to add comments but cant edit
• Comments are not in any order	• Comments are usually in reverse chronological (order that does not remain fixed) order.
• Several authors	• Single author
• Less reliable as anyone can put their own version of information	• It is more reliable as the authors credentials are known

Wifi	Bluetooth
It is used for wide range of communication	Is used for short range of data communication
Wifi is difficult to setup	Easier to set up
More secure	Less secure
Faster data transmission than blue tooth	Slower than wifi
It connects computer, PDA's, printer etc	It connects computers, printers, mouse, microphone, scanner, keyboard etc

Fax:

Fax is a short form of facsimile. It is used to send the exact copy of a document through telephone lines. For sending fax one needs a dedicated fax line and a fax machine.

Advantages:

- Faster than ordinary mailing service
- Fax can be used as legal document
- Can send long documents

Disadvantages

- Higher telephone bills
- Not sure that the right person has received the fax
- Fax is not of good quality /quality of fax deteriorates with the passage of time